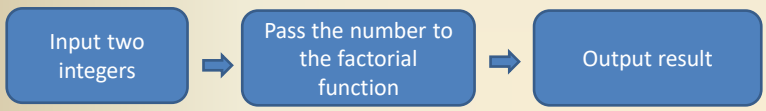
EE1007
Introduction to Computer Science Laboratory

Lab 8
Recursion

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Lab 8.1
P1130291 Combination Calculator (30%)

- Test target: recursion
- You need to use recursive to calculate C(m,n).
- The program don't need to design fool-proof to avoid program errors.
- Flow chart:



Lab 8.1
P1130291 Combination Calculator (30%)

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

factorial function

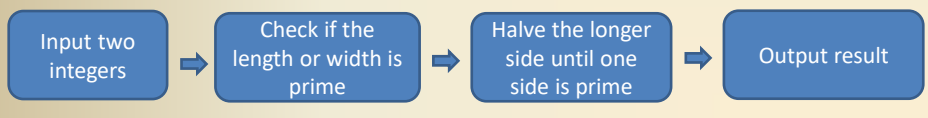
```
long long int factorial(long long int x) {
    if (x == 0)
        return 1;
    else if (x == 1)
        return x;
    else
        return x *= factorial(x - 1);
}
```

Output result

```
cout << "ans: " << factorial(m) / factorial(n) / factorial(m - n);
```


Lab 8.2
P1130292 Rectangle Partitioning (30%)

- Test target: recursion
- You need to write at least one recursive function to calculate the area.
- The program don't need to design fool-proof to avoid program errors.
- Flow chart:



Lab 8.2
P1130292 Rectangle Partitioning (30%)

- Prime function

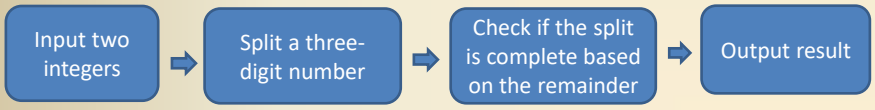
  A prime number is a positive integer greater than 1 that has only two factors: 1 and itself.
  - If the number is divisible by any of the tested numbers, it is not a prime.

- Area function
  - Check if the length or width is prime.
  - Halve the longer side until one side is prime.

```
long long int area(int x, int y) {
    if (prime(x) || prime(y)) {
        return x * y;
    } else {
        return area(max(x, y) / 2, min(x, y));
    }
}
```


Lab 8.3
P1130293 Candy Distribution (30%)

- Test target: recursion
- You need to use recursion, and you cannot use VECTOR or STRING.
- The program don't need to design fool-proof to avoid program errors.
- Flow chart:



Lab 8.3
P1130293 Candy Distribution (30%)

If it's exactly divisible by a after subtraction.

Apply the same for b and c.

```
void remain(int x) {
    if (x - a == 0) {
        counter++;
    } else if (x - a > 0) {
        remain(x - a);
    }

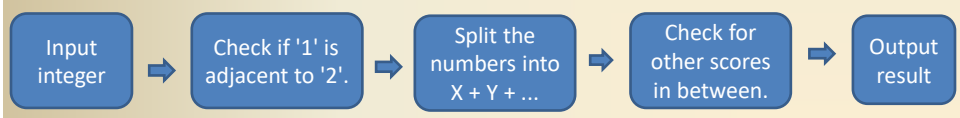
    if (x - b == 0) {
        counter++;
    } else if (x - b > 0) {
        remain(x - b);
    }

    if (x - c == 0) {
        counter++;
    } else if (x - c > 0) {
        remain(x - c);
    }
}
```

If not fully split, then recursively perform x - a.

Lab 8.4
P1130294 Calculate the Score (20%)

- Test target: recursion
- You need to use recursion, and you cannot use VECTOR or STRING.
- The program don't need to design fool-proof to avoid program errors.
- Flow chart:



Lab 8.4
P1130294 Calculate the Score (20%)

```
int calculate(int x, int left, int right) {//left: highest digit, right: lowest digit.
    if (left == right) {
        return 0;
    }

    //1緊接著2
    if (left - right == 1 && x % pow(left + 1) / pow(left) == 1 && x % pow(right + 1) / pow(right) == 2) {
        return 1;
    }

    //判斷並列 X+Y+...
    int symmetry = 0;
    bool b = 0;

    for (long long int i = left; i > right; i--) {
        if (x % pow(i + 1) / pow(i) == 1) {
            symmetry++;
        } else if (x % pow(i + 1) / pow(i) == 2) {
            symmetry--;
        }

        if (symmetry == 0 || symmetry == -1) { //處理2在1前面
            return calculate(x, left, i) + calculate(x, i - 1, right);
        }
    }

    //中間有其他分數 1X2
    if (x % pow(left + 1) / pow(left) == 1 && x % pow(right + 1) / pow(right) == 2) {
        if (allforl(x, left - 1, right + 1))
            return 1;

        return calculate(x, left - 1, right + 1) * 5;
    } else if (x % pow(left + 1) / pow(left) == 1) {
        return calculate(x, left, right + 1);
    }
}
```

//If it's a single digit, the score is 0.

//If it's a two-digit number like '12', the score is 1.

//If symmetry==0 -> "1...2" or "(!(1||2))..."
//If(symmetry==-1) -> "2..."
//Split into X + Y + ...

//Prevent "111...1112"
//calculate (middle score) * 5.
//If "1...(!(1||2))" -> move one position left from the rightmost.

Evaluation

- Lab8.1(30%)
  - 5 testcases, 6% each
- Lab8.2(30%)
  - 5 testcases, 6% each
- Lab8.3(30%)
  - 5 testcases, 6% each
- Lab8.4(20%)
  - 5 testcases, 4% each





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